

Supplemental Information: Noise-Assisted Feedback Control of Open Quantum Systems for Ground State Properties

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I. BOUND FOR $\beta(t)$

For a closed system under unitary dynamics, we have

$$\beta(t) = -A(t) = -\langle [H_d, H_p] \rangle. \quad (1)$$

According to the Cauchy-Schwarz inequality,

$$\frac{1}{2} |\langle [H_d, H_p] \rangle| \leq \Delta H_d \Delta H_p, \quad (2)$$

where $\Delta H = \sqrt{\langle H^2 \rangle - \langle H \rangle^2}$ for an operator H . The variance is upper-bounded by the semi-norm $\|H\|$ of a Hermitian operator H as $\Delta^2 H \leq \|H\|^2/4$. The semi-norm is defined as, $\|H\| = M_H - m_H$, where M_H (m_H) is the maximum (minimum) eigenvalue of H . Defining $n_p = \sqrt{\|H_p\|}$ and $n_d = \sqrt{\|H_d\|}$ and combining with Eq. (2) we find

$$\frac{n_p n_d}{4} \geq \Delta H_d \Delta H_p \geq \frac{1}{2} |\langle [H_d, H_p] \rangle| = -\frac{1}{2} \beta(t), \quad (3)$$

which introduces a lower bound for $\beta(t)$ as $\beta(t) \geq -\frac{n_p n_d}{2}$.

II. BOUND FOR $\Gamma_k(t)$

Consider the non-Markovian master equation, where we define

$$\Gamma_k(t) = -(\langle P_k^\dagger H_p P_k \rangle - \langle H_p \rangle)_t = \text{Tr}[\rho(t)(P_k^\dagger H_p P_k - H_p)], \quad (4)$$

and $\Gamma_k(0) = \lambda_k$. The fidelity between the initial and the final reduced density matrices can be defined as

$$F(t) = \text{Tr}[\rho(0)\rho(t)], \quad (5)$$

as the initial state is a known pure state, i.e., $\text{Tr}[\rho(0)^2] = 1$. Defining $O_k = P_k^\dagger H_p P_k - H_p$ leads to

$$|\Gamma_k(t) - \Gamma_k(0)| \approx \text{Tr}[(\rho(t) - \rho(0))O_k] \leq \|O_k\| \cdot \|\rho(t) - \rho(0)\|_1, \quad (6)$$

where $\|\cdot\|$ is the operator norm (largest singular value), and $\|\cdot\|_1$ is the trace-norm. The trace distance is bounded by

$$\|\rho(t) - \rho(0)\|_1 \leq 2\sqrt{1 - F(t)}. \quad (7)$$

Combining the two equations, we get a bound for the decay operators as

$$|\Gamma_k(t) - \Gamma_k(0)| = |\Gamma_k(t) - \lambda_k| \leq 2 \|O_k\| \cdot \sqrt{1 - F(t)}, \quad (8)$$

where the fidelity at short times can be calculated asymptotically (see Appendix. III).

However, a weaker but easily obtained bound can also be found using the Cauchy-Schwarz inequality for operators $|\text{Tr}(A^\dagger B)|^2 \leq \text{Tr}(A^\dagger A)\text{Tr}(B^\dagger B)$. This leads to

$$|\text{Tr}[\rho(t)(P_k^\dagger H_p P_k)]| \leq \sqrt{\text{Tr}[\rho(t)^2]} \|H_p\|_2, \quad (9)$$

which gives us the final inequality as

$$\begin{aligned} |\Gamma_k(t)| &\leq |\text{Tr}[\rho(t)(P_k^\dagger H_p P_k)]| + \|H_p\| \\ &\leq \sqrt{\mathcal{P}(t)} \|H_p\|_2 + \|H_p\| \\ &\leq 2 \|H_p\|_2, \end{aligned} \quad (10)$$

where $\mathcal{P}(t) = \text{Tr}[\rho(t)^2]$ is the purity of the final state.

III. FIDELITY FOR THE SHORT-TIME QUANTUM DECAY

The bound defined in Eq. (8) requires knowledge of the fidelity $F(t)$, and here we obtain an analytic expression for $F(t)$ in the short-time limit [1]. Its first derivative reads

$$\begin{aligned} \frac{d}{dt} F(t) &= \text{Tr}[\rho(0)\dot{\rho}(t)] \\ &= -i\text{Tr}\{\rho(0)[H(t), \rho(t)]\} \\ &\quad + \sum_{k \in \mathcal{K}} \Gamma_k(t) \{\text{Tr}[\rho(0)P_k \rho(t)P_k^\dagger] - \text{Tr}[\rho(0)\rho(t)]\}. \end{aligned} \quad (11)$$

At $t = 0$, $F'(t)$ reduces to

$$\begin{aligned} F'(0) &= \lim_{t \rightarrow 0} \frac{d}{dt} F(t) \\ &= -\sum_{k \in \mathcal{K}} \Gamma_k(0) \{\text{Tr}[\rho(0)^2] - \text{Tr}[\rho(0)P_k \rho(0)P_k^\dagger]\} \\ &= -\sum_{k \in \mathcal{K}} \Gamma_k(0) \Delta P_k^2, \end{aligned} \quad (12)$$

with the variance of the operator $\Delta P_k^2 = \langle P_k^2 \rangle - \langle P_k \rangle^2$ and $\Gamma_k(0) = \lambda_k$. Hence, in a short-time asymptotic expansion, the

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fidelity can be written as

$$\begin{aligned} F(t) &= 1 + F'(0)t + \mathcal{O}(t^2) \\ &\approx 1 - t \sum_{k \in \mathcal{K}} \lambda_k \Delta P_k^2. \end{aligned} \quad (13)$$

IV. LYAPUNOV CONTROLLED DECOHERENCE-FREE SUBSPACE

The decoherence-free subspace (DFS) method was introduced as an alternative method for error-free quantum computation [2]. This is an important concept, as the states in DFS evolve unitarily, even in the presence of decoherence. The Markovian dynamics representing the noise in the NISQ hardware is governed by Eq. (4) of the main text with the learned positive error rates, $\{\lambda\}$. Here, we add an extra Lyapunov control field $\gamma(t)$ and a control Hamiltonian H_n to cancel out the dissipative part of the master equation. With the additional controls, the master equation becomes

$$\frac{d\rho}{dt} = -i[H_p + \beta(t)H_d + \gamma(t)H_n, \rho] + \mathcal{D}_{\text{int}}[\rho], \quad (14)$$

where $\mathcal{D}_{\text{int}}[\rho]$ is the stochastic Pauli noise channel of the NISQ device. For the new controlled Hamiltonian, Eq. 5 (main text) reduces to

$$\langle \dot{H}_p \rangle = \beta(t) \langle i[H_d, H_p] \rangle + \gamma(t) \langle i[H_n, H_p] \rangle + \text{Tr}(\mathcal{D}_{\text{int}}[\rho]H_p). \quad (15)$$

Making a connection with [3], we choose

$$\begin{aligned} \gamma(t) &= -\frac{\text{Tr}(\mathcal{D}_{\text{int}}[\rho]H_p)}{\langle i[H_n, H_p] \rangle}, \\ \beta(t) &= -\langle i[H_d, H_p] \rangle, \end{aligned} \quad (16)$$

such that $\langle \dot{H}_p \rangle \leq 0$. There should always exist a $\gamma(t)$ unless $\langle i[H_n, H_p] \rangle = 0$. This approach can be called the Lyapunov-controlled decoherence-free subspace (LC-DFS).

Here we discuss the limitations of the LC-DFS compared to those of NAFQA. First, the choice of additional Lyapunov control Hamiltonians is restricted due to the condition $\langle i[H_n, H_p] \rangle \neq 0$. Second, the commutator $[H_n, H_p]$ may contain higher-order terms, which will be difficult to implement on quantum computers. For example, for the spin glass Hamiltonian considered in the main text, $H_p = \sum_{m < n} J_{mn} Z_m Z_n + \sum_l h_l Z_l$, and a two-body control Hamiltonian $H_n = \sum_{i < j} (Y_i Z_j + Z_i Y_j)$, the commutator $[H_n, H_p]$ contains one-body X , two-body XZ, ZX , and three-body XZZ, ZZX operators. In this case, implementing the three-body operators on quantum computers is not straightforward, which further restricts the choice of control Hamiltonians and the application of the LC-DFS approach.

V. AMPLITUDE AND PHASE DAMPING NOISE CHANNELS

In the main text, we have considered the implementation of NAFQA with the stochastic Pauli noise (unital) channels.

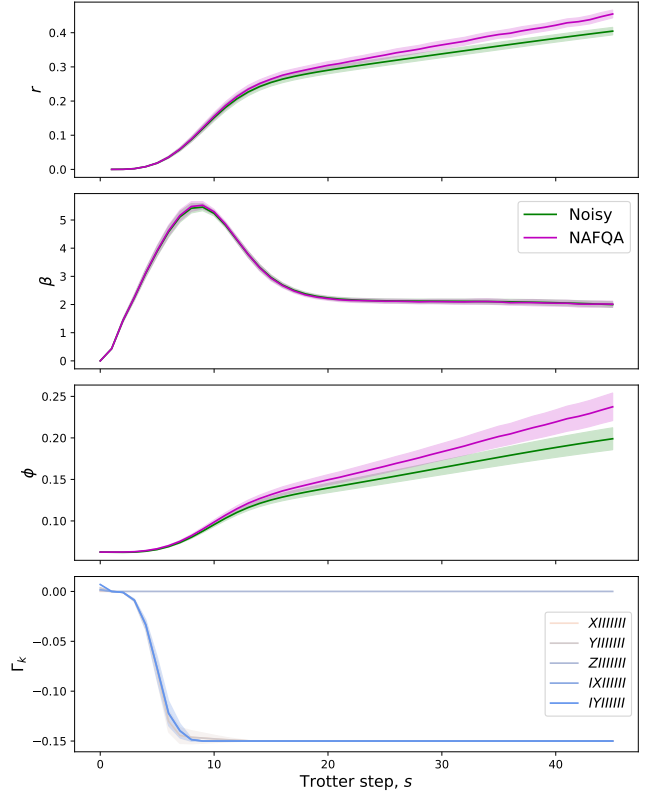


FIG. 1. **8-qubit spin glass Hamiltonians.** The performance of NAFQA against the noisy FQA with error probabilities given in Table I. The mean and standard error (a) approximation ratio r , (b) control field β , and (c) success probability ϕ over 25 instances (seeds 0 to 24) are illustrated. The numerical simulations are carried out with a time step of $dt = 0.01$, yielding NAFQA results from 5000 trajectories. (d) Error probabilities $\Gamma_k(t)$ (for $k \in \{X, Y, Z\}$) that can take negative values.

However, one can also study the behavior of the NAFQA algorithm beyond the stochastic Pauli noise channel, for instance, with the amplitude and phase damping channels, which are widely considered in OQS dynamics [4]. The GKSL equation corresponding to the local amplitude and phase damping is given by

$$\frac{d\rho}{dt} = -i[H, \rho] + \mathcal{D}_r[\rho]. \quad (17)$$

The Lindblad dissipator takes the form

$$\begin{aligned} \mathcal{D}_r[\rho] &= \sum_{k=1}^N \lambda_k \left(s_k^{(-)} \rho s_k^{(+)} - \frac{1}{2} \{ s_k^{(+)} s_k^{(-)}, \rho \} \right) \\ &\quad + \sum_{k=1}^N \frac{\lambda_k}{4} (Z_k \rho Z_k - \rho), \end{aligned} \quad (18)$$

which is a combination of the amplitude damping noise with decay rates λ_k and the phase damping noise with rates $\lambda_k/4$. Here, $s_k^{(\pm)} = \frac{1}{2} (X_k \pm iY_k)$, and $s_k^{(+)} s_k^{(-)} = |1\rangle\langle 1|_k$ corresponds to the k^{th} qubit. The condition for the single-qubit dephasing

channel follows a procedure similar to that of the stochastic Pauli noise (Eq. (7) of the main text). However, the Lyapunov condition for the amplitude-damping channel takes the form

$$\frac{d}{dt} \langle H_p \rangle = \beta(t) \langle i[H_d, H_p] \rangle + \sum_{k \in \mathcal{K}} \Gamma_k(t) \left(\langle s_k^{(+)} H_p s_k^{(-)} \rangle - \frac{1}{2} \langle \{H_p, s_k^{(+)} s_k^{(-)}\} \rangle \right) \leq 0. \quad (19)$$

The amplitude and phase damping channels can be implemented on quantum circuits with or without the help of ancilla qubits [5, 6]. Similarly to the feedback-based noise-assisted approach discussed in the Results, one can engineer the time-dependent decay rates based on the feedback law to use the amplitude and phase damping channels as a resource to assist quantum simulations.

VI. CONDITIONS (C1) AND (C2)

Both conditions introduced in the main text are verifiable from the operator structure alone, without knowledge of the eigenvalues or eigenstates of H_p . Condition (C1) is checked by computing the rank of the Lie algebra generated by $\{iH_p, iH_d\}$ via repeated commutators [7, 8]. Condition (C2) reduces to verifying $[P_k, H_p] \neq 0$ for at least one $k \in \mathcal{K}$, which holds whenever the noise model contains Pauli operators outside the stabilizer group of H_p . Both conditions are satisfied for all MaxCut and spin-glass instances studied here.

Within the closed-system QLC framework, Ref. [9] establishes that asymptotic convergence to the ground state is guaranteed under four conditions: (1) H_p has no degenerate eigenvalues; (2) H_p has no degenerate eigenvalue gaps; (3) $\langle q_j | H_d | q_i \rangle \neq 0$ for all $i \neq j$, where $|q_i\rangle$ and $|q_j\rangle$ are the eigenvectors of H_p ; and (4) $E_p(|q_0\rangle) < E_p(|\psi(0)\rangle) < E_p(|q_1\rangle)$. Conditions (1)–(3) jointly imply condition (C1). Condition (C1) alone therefore restricts $\mathcal{S}$ via the unitary part of the dynamics, in direct analogy with the closed-system case.

Condition (C2) acts through the dissipative term and further excludes excited eigenstates from $\mathcal{S}$. Importantly, condition (C2) plays the role of condition (4) of Ref. [9] without imposing any constraint on the initial state: the additional dissipative term $\sum_k C_k(t)^2$ in the NAFQA energy derivative acts as an independent filter that removes excited states from $\mathcal{S}$, a mechanism absent in the unitary-only FQA dynamics. NAFQA therefore operates under a strictly weaker set of initialization assumptions than closed-system FQA [9].

Whether (C1) and (C2) jointly guarantee $\mathcal{S} = \{|q_0\rangle\langle q_0|\}$ for general mixed-state dynamics remains an open question [3, 10], as the NAFQA trajectory is generically mixed due to the dissipators, and a complete characterization of the invariant set under open-system dynamics requires further analysis. This is left for future work.

TABLE I. **Synthetic sparse Pauli-Lindblad noise model for $N = 8$ qubits.** The model comprises 20 weight-1 and 40 weight-2 Pauli error terms, generated with a fixed seed (seed = 42) from a log-uniform distribution on $[4 \times 10^{-6}, 3.4 \times 10^{-3}]$, consistent with the sparsity and magnitude of the $N = 5$ device-characterized noise model shown in the main text. The numerical simulations are performed with the error probabilities rescaled by a multiplicative factor of 10, i.e., $\tilde{\lambda}_p = 10 \lambda_p$.

Pauli string	Error probability (λ_p)	Pauli string	Error probability (λ_p)
XIIIIIII	8.6610×10^{-6}	IXIYIII	1.1744×10^{-4}
YIIIIIII	3.6316×10^{-4}	IYIIZIII	1.1175×10^{-5}
ZIIIIIII	9.5962×10^{-5}	IZIIZIII	4.3840×10^{-4}
IXIIIIII	1.8108×10^{-4}	IZIYYII	8.1103×10^{-5}
IYIIIIII	6.9673×10^{-4}	IZIIZII	5.2267×10^{-5}
IIXIIIIII	2.8934×10^{-4}	IZIIXXI	3.0571×10^{-5}
IYYIIIIII	1.6739×10^{-4}	IYIIIIIZ	2.8082×10^{-4}
IIZIIIIII	1.7386×10^{-4}	IZIIXIX	4.5916×10^{-5}
IIIXIIII	3.1078×10^{-5}	IIXYIII	7.2248×10^{-6}
IIYIIIIII	4.9242×10^{-6}	IIZZIIII	8.8664×10^{-6}
IIIZIIII	7.6101×10^{-5}	IIZIZIII	2.6294×10^{-3}
IIIXIIII	1.7009×10^{-5}	IIZIYYII	1.8352×10^{-3}
IIIIYIII	6.2923×10^{-5}	IIZIIXXI	4.4853×10^{-4}
IIIIIXII	1.2648×10^{-3}	IIZIYYI	2.4038×10^{-5}
IIIIIXI	1.9381×10^{-5}	IIZIIXIX	2.7617×10^{-3}
IIIIIIYI	5.9272×10^{-6}	IIXXXIII	7.6445×10^{-4}
IIIIIIZI	2.6690×10^{-5}	IIZZZIII	5.0365×10^{-4}
IIIIIIIX	2.8981×10^{-5}	IIXXIYYI	8.2876×10^{-5}
IIIIIIYY	3.4761×10^{-4}	IIXXIZII	2.5094×10^{-5}
IIIIIIIZ	1.7133×10^{-4}	IYYIYYII	7.6635×10^{-6}
XIYIIIII	7.9146×10^{-4}	IYYIIZ	1.7626×10^{-3}
YIZIIIII	3.5327×10^{-4}	IIZIIZII	8.6541×10^{-5}
XIIXIIII	6.2021×10^{-5}	IIIIYIZI	1.5663×10^{-5}
YIIIZIII	9.6977×10^{-4}	IIIIIZI	3.1502×10^{-5}
ZIIZIIII	1.2337×10^{-5}	IIIIIXIY	1.9899×10^{-4}
ZIIIIYII	4.6622×10^{-6}	IIIIIIYYI	1.3180×10^{-5}
XIIIIIIYI	7.3426×10^{-6}	IIIIIIYZI	1.2925×10^{-3}
XIIIIIIIZ	5.2258×10^{-4}	IIIIIIIXZ	6.6693×10^{-4}
IYIYIIII	9.0176×10^{-5}	IIIIIIYY	5.1247×10^{-4}
IYIZIIII	1.1871×10^{-5}	IIIIIIIZX	7.3764×10^{-5}

VII. ADDITIONAL NUMERICAL SIMULATIONS

8-qubit spin glass Hamiltonians. To further validate the performance of NAFQA, we examine its behavior on spin-glass instances with $N = 8$ qubits. The problem Hamiltonian takes the all-to-all connected form

$$H_p = \sum_{m < n} J_{mn} Z_m Z_n + \sum_l h_l Z_l, \quad (20)$$

driven by the control Hamiltonian $H_d = -\sum_{j=1}^8 X_j$. The corresponding commutator evaluates to

$$i[H_d, H_p] = -2 \left(\sum_{i < j} J_{ij} (Y_i Z_j + Z_i Y_j) + \sum_i h_i Y_i \right). \quad (21)$$

The interaction strengths J_{mn} and local field coefficients h_l are independently sampled from a uniform distribution on $[-1, +1]$, and the statistics are collected over 25 instances (seeds 0 through 24) using 5000 Monte Carlo trajectories per instance. The mean and standard error of the approximation ratio r , the success probability ϕ , and the control field β are displayed in Fig. 1.

We did not characterize the coefficients of the noise model for $N = 8$ and we employ a synthetic sparse Pauli-Lindblad noise model, constructed to mirror the structural properties of the $N = 5$ device-characterized model shown in the main text. Specifically, the model comprises 20 weight-1 and 40 weight-2 Pauli error terms, with the number of terms scaled polynomially from the $N = 5$ baseline as $|\mathcal{K}| \sim \mathcal{O}(N^2)$, consistent with the sparse model assumption $|\mathcal{K}| \ll 4^N - 1$. The error

probabilities $\{\lambda_P\}$ are drawn from a log-uniform distribution on $[4 \times 10^{-6}, 3.4 \times 10^{-3}]$, matching the range and sparsity of the $N = 5$ data, and are listed in Table I. We emphasize that this noise model is not device-characterized and is employed solely for illustrative purposes, to demonstrate the scalability and robustness of the NAFQA algorithm. Noisy simulations are performed with the error probabilities of Table I, scaled by a multiplicative factor of 10. As with the $N = 5$ spin-glass simulations in the main text, this amplification emulates the effect of stronger noise or, equivalently, in the noise-dominated regime established by Proposition 1 of the main text, the accumulation of noise at circuit depths approximately 10 times larger than those simulated.

Five Lyapunov-controlled decay rates Γ_k ($k = 1, 2, 3, 4, 5$) are adaptively updated at each Trotter step, with their magnitudes bounded by a threshold of -0.15 . The results confirm that NAFQA retains its ability to accelerate convergence toward the ground-state properties compared with the noisy FQA evolution, providing evidence of the algorithm's scalability to moderately large system sizes.

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